Numerical Analysis (2627-MA385): **Class Test** ANS With answers

When:	15:00 Thursday, 29 October 2026 (CSB-1005 and AC201).
Duration:	40 minutes.
Instructions:	Answer all questions in the answer book provided
Don't forget:	to write your name and ID number on the answer book.
Rules:	No notes, books or log tables allowed. Only basic calculators permitted. Don't sit beside anybody else.

Q1. (a) State the *Secant Method* for solving the nonlinear equation $f(x) = 0$.

Answer: Choose two points x_0 and x_1. Then set

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \qquad \text{for } k = 1, 2, 3, \ldots$$

(b) Give a derivation (or careful justification) for this method.

Answer: Among the acceptable answers is that we'll take x_{k+1} to be the zero of the line joining the points $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$. If f is roughly linear in this region, then that should be a good approximation for the solution. The formula for the line is $y - f(x_k) = \frac{f(x_k)-f(x_{k-1})}{x_k - x_{k-1}}(x - x_k)$. If $x = x_{k+1}$ gives $y = 0$, then this gives us $-f(x_k) = \frac{f(x_k)-f(x_{k-1})}{x_k - x_{k-1}}(x_{k+1} - x_k)$. Rearrange to finish.

(c) We wish to use the Secant Method to estimate $\tau = \sqrt[3]{4}$ numerically by solving $f(x) = 0$ in $[a, b]$ for some suitably chosen f, a and b. Suggest suitable choices of f, a, and b for this problem, and show that f has a zero in $[a, b]$.

Answer: We can solve $f(x) = x^3 - 4 = 0$ for $[a, b] = [1, 2]$. Then $f(a)f(b) = f(1)f(2) = (-3)(4) = -12 < 0$. Therefore f changes sign on $[1, 2]$. Consequently, it must have a zero in $[1, 2]$.

(d) Take $x_0 = a$ and $x_1 = b$. use 3 steps of the Secant Method method to estimate $\sqrt[3]{4}$. (That is, compute the estimates x_2, x_3 and x_4).

Answer: You should find, to 6 digits, $x_2 = 1.42857$, $x_3 = 1.550459$, and $x_4 = 1.59142$.

Q2. (a) State *Newton's method* for solving the nonlinear equation $f(x) = 0$ *for some* $x \in [a, b]$.

Answer: Choose an $x_0 \in [a, b]$ and set

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \qquad \text{for } k = 0, 1, 2, \ldots. \tag{1}$$

(b) Give a derivation of Newton's Method, and the Newton Error formula, using a truncated Taylor Series.

Answer: We start with the Truncated Taylor Series

$$f(\tau) = f(x_k) + (\tau - x_k)f'(x_k) + \frac{1}{2}(\tau - x_k)^2 f''(\eta_k), \qquad \text{for some } \eta_k \in (\tau, x_k).$$

Since $f(\tau) = 0$, this gives the following expression for τ:

$$\tau = x_k - \frac{f(x_k)}{f'(x_k)} - \frac{1}{2}(\tau - x_k)^2 \frac{f''(\eta_k)}{f'(x_k)}. \tag{2}$$

If we neglect the $(\tau - x_k)^2$ term, and denote the approximation of τ as x_{k+1}, we get the method in (1). If we don't, and substitute (1) into (2) we get the Newton Error Formula:

$$\tau - x_{k+1} = -\frac{1}{2}(\tau - x_k)^2 \frac{f''(\eta_k)}{f'(x_k)}.$$

(c) Generally, what do you expect the Order of Convergence to be for Newton's Method, and how is that justified by the Newton Error Formula? *Note: you don't have to prove convergence, just give a justification.*

Answer: We expect it to be second order because, if $x_k \to \tau$ as $k \to \infty$, then $\eta_k \to \tau$. So we take $\mu = \frac{1}{2}\frac{f''(\tau)}{f'(\tau)}$ and $\dfrac{|\tau - x_{k+1}|}{(\tau - x_k)^2} \to \mu$.

(d) Suppose we take $f(x) = x^2$, and apply Newton's method to solve $f(x) = 0$ with $x_0 = 1$. What order of convergence does one observe in practice? Does this agree with your answer from Part (c)? If not, offer an explanation as to why.

Answer: For this problem, Newton's method reduces to $x_{k+1} = x_k - (x_k^2)/(2x_k) = x_k/2$. Clearly this converges linearly to the expected solution $\tau = 0$. So we don't get second-order convergence. This is because, tacitly, the analysis assumed $f'(\tau) \neq 0$.

Q3. (a) State *Euler's method* for solving initial value problems:

$$y(t_0) = y_0 \qquad \text{and } y'(t) = f(t, y) \text{ for } t > t_0.$$

Answer: Euler's method is $y_{i+1} = y_i + hf(t_i, y_i)$, for $i = 0, 1, \ldots, n-1$.

(b) Show how Euler's method can be motivated by using a truncated Taylor Series.

Answer: Taylor series for $y(t_{i+1})$ about t_i:

$$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i)y'(t_i) + \frac{(t_{i+1} - t_i)^2}{2}y''(\eta_i),$$

for some $\eta_i \in [t_i, t_{i+1}]$. Using $h = t_{i+1} - t_i$, and that, from the IVP, $y'(t_i) = f\big(t_i, y(t_i)\big)$, giving

$$y(t_{i+1}) = y(t_i) + hf\big(t_i, y(t_i)\big) + \frac{h^2}{2}y''(\eta_i).$$

Neglecting the $\mathcal{O}(h^2)$ term, and denoting the resulting approximation for $y(t_i)$ as y_i, we get the method.

(c) Give an example of an Initial Value problem for which Euler's method would give the exact solution (i.e., with no error). Justify your answer with the Taylor Series you used to motivate Euler's method.

Answer: We need a problem for which the solution has $y''(t) \equiv 0$ for all t. Take, for example $y(t) = t$, so $y'(t) = 1$, and $y''(t) = 0$. Now we just need a problem for which this is a solution. Acceptable options include taking $f(t, y) = 1$, or (equivalently) $f(t, y) = y/t$, along with the initial condition (for example) $y_0 = 1$ at $t_0 = 1$.